



# Towards Deriving Organizational Key Performance Indicators in a Metaverse Workplace: A Systematic Literature Review

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## ABSTRACT

Human resources of several businesses worldwide are already active in metaverses. This has led Human Resource Management (HRM) professionals exploring and applying new methods and practices to manage this technological transformation in the workplace. However, despite the growing volume of research on metaverse work environments, there is still limited research on avatar performance management. The emergence of metaverse applications in the workplace brings new key performance indicators (KPIs) requirements that are different from those in conventional work environments. This article aims to fill this gap by identifying the key metrics for measuring and managing avatar performance. For this purpose, a systematic literature review is carried out based on the PRISMA methodology. In total, 357 articles were screened, from which 31 met the predefined inclusion criteria and were reviewed based on their risk of bias. As a result, the required KPIs suitable for a metaverse workplace are proposed. By contributing to knowledge on performance in virtual work environments, this article extends and advances theory and research on the Strategic Human Resource Management of metaverse workplaces.

## CCS CONCEPTS

• Reengineering; • Social and professional issues; Employment issues;

## KEYWORDS

Metaverses, Human Resource Performance, Performance Appraisal Measurements, Key Performance Indicators (KPIs)

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## 1 INTRODUCTION

The Fourth Industrial Revolution has brought about significant changes to the workplace, including the emergence of metaverse-based teleworking offices. With an estimated 10% of workers using virtual workspaces by 2025, measuring employee performance in the metaverse is becoming increasingly important. However, traditional performance measurement systems may not be suitable for this new environment, which imposes new approaches to measuring performance compared to the financial and accounting indicators used by organizations. As such, it is essential to identify critical performance indicators for Avatar performance in the Metaverse workplace that can be implemented by HR executives to manage the technological transformation of the workplace effectively. This study aims to fill this gap by identifying and determining important indicators that influence the success of Avatars in the metaverse workplace through a Systematic Literature Review. The findings of this study provide a foundation for designing an effective Performance Management Model, which would enable organizations to manage better the impact of the metaverse workplace on employee performance. This paper utilizes the PRISMA and ROBIS methodologies to minimize bias in the review and develop a conceptual model for further research. The increasing interest in the topic of 'employee performance in the metaverse' detected in Google Scholar highlights the need for further research in this area. Overall, this study aims to organize scientific knowledge about the impact of the metaverse workplace on employees' performance and drive a comprehensive list of KPIs for Avatars required in a metaverse workplace [1-10].

## 2 LITERATURE REVIEW

HR management plays a crucial role in the process of employee performance appraisals [11]. Performance management is designed to help organizations achieve their strategic objectives by maintaining and developing employees' performance [12]. However, developing an effective appraisal system can be challenging, particularly in the context of a new technological work environment. Clear and measurable performance indicators are considered essential in ensuring the accuracy and success of the system [13, 14]. Performance measurement starts with identifying performance indicators [15], which has become a recent concept in business management. Traditionally, KPIs included only financial and accounting indicators [6]. However, with the rapidly changing societal and technological environment, multi-dimensional models of KPIs have emerged [16]. For example, in the metaverse workplace, one performance measurement method for employees is rank-based reviews based on

financial metrics such as sales performance. With the current transformation of companies to the metaverse, sales may decrease, and workers may fear layoffs [7]. As such, companies require changes to their employee performance appraisal systems to suit their new structure [17, 18].

In the metaverse environment, employees manifest as programmable digital bodies known as avatars [19]. While traditional employees are people hired by organizations to perform specific jobs or tasks, an employee's avatar is a digital representation of that employee, often used in virtual environments or online platforms. However, there are challenges in terms of the technology and hardware required to create and run employee avatars in the metaverse, which can be costly and time-consuming for organizations to implement. Additionally, avatars allow for a heightened sense of self-expression, as users can customize their avatar's features to accurately reflect their personal expression [20, 21].

It is worth considering the Proteus effect, which refers to the significant psychological impact that a person's virtual representation (avatar) can have on their behavior and learning in virtual environments. Embodied experiences as avatars directly affect human behavior, which can then transfer to the real world [22]. Furthermore, a metaverse workplace has an impact on employees' flexibility, autonomy, and other factors such as technostress, techno-spatial intrusion, techno-addiction, sexual harassment, and attacks, additional required skills, cyber security, and privacy issues, technical limitations, and communication challenges [23-27]. All these factors affect employee-level outcomes such as productivity, performance, and job satisfaction. Armstrong and Baron [28] stated that measurement is the core of performance management as it generates feedback and identifies the good and bad things within the organization [29], such as the challenges and issues that employees face in the metaverse workplace. Therefore, to portray performance, productivity, and satisfaction levels accurately, there is a need to develop several key performance indicators for each employee issue in the metaverse workplace.

## 2.1 The impact of a metaverse workplace on employees' performance

The impact of a metaverse workplace on employees' performance is a complex issue that requires consideration of various factors [23-25, 30-38, 40-42]. Communication challenges, training needs, employee confidence, flexibility and autonomy, security and safety, rate of completed work, and time evaluation of work are all aspects of employee performance that may be affected by working in a metaverse environment [23, 33-38]. Employers and managers must understand these factors to better support their employees and improve their performance in this new working environment [23, 30, 34, 37].

Effective communication channels must be established to improve communication in a metaverse workplace, considering interaction, geographic and temporal coordination, and technology limitations [25]. Training needs must be considered to support employees in a metaverse workplace, including additional skills such as multi-tasking, appearance and behavioral realism, and avatar design [23, 31-33]. Employers must provide the necessary training and support to improve employee confidence, particularly in the face

of the risks in the workplace [23, 34]. Flexibility and autonomy are advantages of a metaverse workplace, but managing security and safety issues can lead to stress for employees, which can hurt their performance [23, 35-37]. Security and safety concerns, including techno-stress, techno-spatial intrusion, techno-addiction, sexual harassment, attacks, cyber security, and privacy issues, are present in a metaverse workplace and can impact employee well-being and productivity [23, 24]. Avatar appearance can impact customer credibility and job productivity, affecting the rate of completed work [33]. An employee's anxiety and training needs can negatively impact the time required to complete a job, and time performance metrics can be useful in evaluating employee performance [23, 38, 40, 41].

The current performance management system is inadequate for measuring and improving the performance of avatars in the metaverse workplace [23]. Therefore, a new approach to performance measurement is proposed, based on employee-avatar job performance, productivity, and job satisfaction, taking into account avatar development and the new metaverse workplace [28]. This approach can be used to develop the Avatar Performance Appraisal System (A), which includes job performance (B), job satisfaction (C), and employee productivity (D) indicators [28] (see Figure 1). HR professionals can use these indicators to address employee issues in the metaverse workplace and develop their performance following the next steps of their performance appraisal system (feedback and development plan, review and improvement) [28].

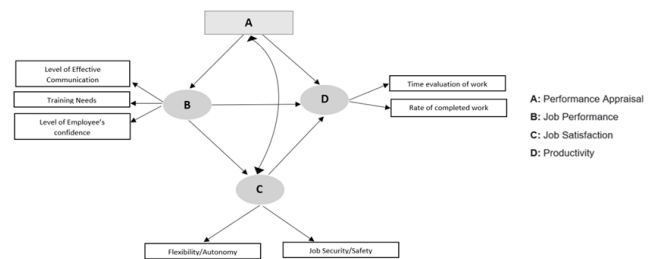


Figure 1: Avatar-Metaverse-Performance Conceptual model diagram

In conclusion, a metaverse workplace presents unique challenges that require a new approach to performance management [23]. Employers must understand the various factors that can impact employee performance and take steps to support their employees effectively [23, 30, 34, 37]. Further research is needed to fully understand the impacts of a metaverse workplace and develop robust and validated metrics for measuring avatar performance [23, 28, 42]. It is also essential to consider factors such as age, disabilities, and culture, which may play a significant role in employee performance levels when implementing metaverse technologies [23, 36, 37, 42].

## 2.2 KPIs

Performance indicators can cover various aspects such as quality, finance, flexibility, employee satisfaction, safety, community/environment, and learning/growth [40-51]. In virtual environments, several KPIs, including usage, occupation time, user interaction, and eye-tracking, should be considered [52]. However, there may be trade-offs between performance metrics, and improving

one metric may result in a decrease in another [20]. Much research has examined the relationship between job satisfaction, productivity, and performance [44, 54, 55]. Engaged and satisfied employees tend to be top performers who are committed to the organization and serve customers better, leading to increased profitability [44]. In conclusion, job performance, productivity, job satisfaction, and performance appraisal are variables with correlational relationships. By examining the main issues of avatars in the metaverse, HR professionals can determine the KPIs of an Avatar Performance Appraisal System, which include job performance, job satisfaction, and employee productivity [28]. Through these indicators, HR professionals can address employee issues in the metaverse workplace and develop their performance following the next steps of their performance appraisal system.

**2.2.1 Explaining the proposed indicators.** A: Performance Appraisal System: The proposed Key Performance Indicators (KPIs) for the performance appraisal system in the metaverse environment are critical in assessing employee performance, identifying areas for improvement, and ensuring that employees are motivated, engaged, and productive. KPIs can include measures such as job performance, satisfaction, and productivity [28, 56].

B: Avatar's Performance: Due to the additional skills required and technical limitations in communication in the metaverse environment [23, 25], HR managers must measure job performance by examining the level of effective communication in the virtual workplace, training needs, and employee confidence [28]. This could also include tracking employee interactions with the virtual environment and eye-tracking for focus validation [52].

C: Avatar's Satisfaction: This refers to the avatar's level of satisfaction with the metaverse environment and its work within it. HR managers can identify the level of job satisfaction by measuring workers' job security, stress, autonomy, and flexibility [56–59]. Employee satisfaction measurement at work requires careful observation of their opinion, feelings, and experience within the working environment.

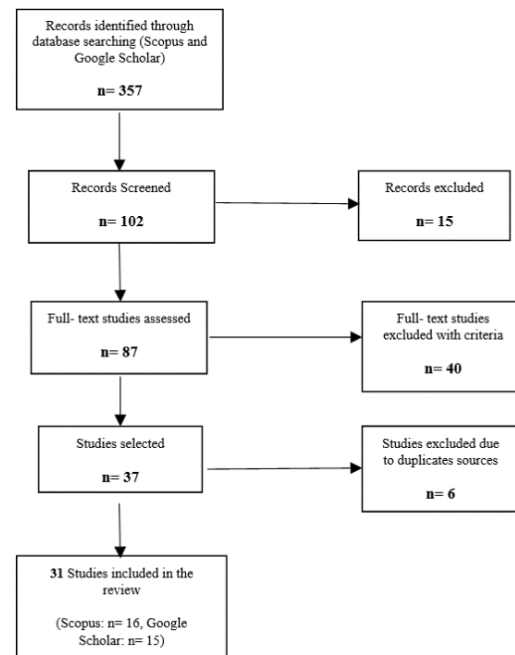
D: Avatar's Productivity: This refers to the avatar's ability to complete tasks and achieve goals in the metaverse environment. Measuring productivity can include tracking the number of tasks completed and the time it takes to complete them [60]. Productivity can also be measured by looking at the rate of effective work and time evaluation of employee work [60].

It is important to note that while these proposed KPIs can help in measuring employee performance in the metaverse environment, there can be compromises between performance metrics. For instance, improving one metric may result in a decrease in another. Thus, HR managers must take a comprehensive approach to measuring employee performance to ensure that it is balanced and aligns with the organization's strategic objectives.

### 3 METHODOLOGY

One issue with the performance measurement literature is its diversity, leading authors to concentrate on different aspects of performance measurement system design. This study conducted a systematic literature review to map the scientific knowledge about the impact of metaverse teleworking offices on employees, focusing on avatars' performance appraisal measurements [61, 62]. This

method was chosen because it helps to synthesize academic literature accurately and reliably. In addition, such a review integrates findings from unrelated studies on a common issue, enabling the contrast and comparison of results and the identification of areas that lack adequate scientific research, which would become areas of future study [61, 62]. The systematic literature review was conducted by the preferred reporting items of the Prisma approach [50, 63]. The PRISMA approach entails an evidence-based checklist of 27 items and a four-phase flow diagram (see Figure 2) [63].



**Figure 2: Avatar-Metaverse-Performance Study selection workflow**

The checklist items included evidence that not reporting the thing was associated with an increased risk of bias or where it was clear that information was necessary to appraise the reliability of a review. (Checklist items were included if required to ascertain the reliability of the review or if their exclusion would have increased the risk of bias.) PRISMA is not intended to be a quality assessment tool but aims to ensure clarity and transparency of the systematic literature reviews [9]. As such, the article examined the variables and their possible inter-relationships through two main aims. First, were organizing the existing scientific data about the impact of metaverse workplace on employees' performance as Avatars and, secondly, the improvement of avatar's performance appraisal through the KPIs [61, 62].

Existing research included in the study was selected according to specific criteria. Only peer-reviewed studies were considered from all years, with abstracts and full texts [64, 65]. Scopus and Google Scholar, reputable international databases of scientific publications, were the sources for this extensive research (Search terms were carefully selected to optimize several results). Choosing search terms

is essential for a more effective search that may yield the highest results. The research question provided a list of suitable keywords for a productive search, as many as necessary to cover the investigation. This study used the combination of keywords: Performance Appraisal AND hrm; key performance indicators AND hrm; meta-verse office AND employee performance; job performance AND key performance indicators; job satisfaction AND key performance indicators; employee productivity AND key performance indicators [66, 67].

The article selection process started with 357 papers. After eliminating the repetitions, 102 articles remained for the eligibility process: which entailed full-text analysis to determine whether they met the above inclusion criteria. Studies on metaverse teleworking offices in fields other than Human Resources were excluded, as were those which examined technical parts of the technology, for example, their manufacture or programming systems. The research thus only included studies related to the impact of metaverse workplace on Human Resources and employee performance appraisal. Subsequently, a definitive directory of studies relevant to the research objectives was established through a full-paper analysis of the initial list. The final list included 31 articles registered in an Excel sheet [64, 65].

## 4 RESULTS AND MAIN FINDINGS

The systematic literature review and risk of bias (PRISMA) assessment revealed the need for new workplace key performance indicators for measuring avatar performance in metaverse environments, particularly in relation to avatar performance, productivity, and satisfaction. However, to develop more robust and validated metrics for measuring avatar performance, further research is necessary. The reviewed studies covered various aspects of organizational communication, remote work, employee well-being, performance measurement, virtual reality, employee morale, industry 4.0, and crisis management.

As the implementation of new technologies and ways of working can affect employee well-being, organizations must consider their impact on employee satisfaction, performance, and productivity. The selection of key performance indicators is crucial for effective performance measurement. The systematic review also revealed the importance of managing change and supporting employees as they adapt to new ways of working. Measuring KPIs for avatars based on factors such as job satisfaction, performance measurement, and productivity can help improve employee experience and efficiency while addressing potential security and privacy issues. Table 1 identifies various sources of bias or errors that may deviate the study's results from the truth. The reviewed studies exhibit low, unclear, or high-risk bias, indicating the effectiveness and reliability of the study's outcomes.

For instance, randomly selected data without specification reduces selection bias and produces optimal results [10]. However, when the bias on the binding of outcomes is unclear, it is impossible to know whether the assessor knew of the possible results or intervention, making the study's outcomes less reliable. Lastly, high-risk bias indicates a high chance of the outcomes being unreliable since the bias has a high probability of changing the results. When the reporting bias is high, it means that the assessor or researcher did

not report a lot of information found from the study's data, and thus the outcomes of the study are not adequate to rely on. Since most of the studies' data have a low-risk bias, the outcomes observed in the sources have high chances of being true and can be relied on in the literature review. Therefore, it is crucial to consider bias when conducting systematic reviews and to employ appropriate methods to minimize bias to increase the effectiveness and reliability of the study's outcomes.

## 5 DISCUSSION & CONCLUSIONS

The emergence of the Metaverse era presents significant opportunities and challenges for organizations across various industries. The need for new workplace key performance indicators (KPIs) to measure avatar performance in metaverse environments has been highlighted through a systematic literature review. The application of employee performance measurement systems in a metaverse workplace tends to rely on three basic criteria: employee-avatar job performance, productivity, and job satisfaction. These would be the main determinants of work effectiveness.

The current performance management system may not accurately evaluate employee performance, leading to inaccurate feedback and incorrect decisions about promotions and salaries. Moreover, the current system may be outdated and not relevant to the current work environment, subjective, leading to a lack of fairness and consistency in the evaluation of employees, and may only focus on technical skills and ignore other important aspects of an employee's performance, such as teamwork, leadership, and communication. Furthermore, the current system may not provide enough feedback to employees, limiting their ability to improve their performance.

By implementing a new metaverse performance appraisal management system, organizations can address these issues and ensure a more accurate, fair, and comprehensive evaluation of employee performance. Businesses would venture to implement the integrated performance measurement system only if they are convinced that the benefits, they will achieve by implementing such a system would be more than the cost and effort they put into designing the system. This can only be established by developing the right KPIs.

The field is currently under-explored, and further research is necessary to assist companies in coping with the effects of the metaverse technological metamorphosis. It is also essential to consider factors such as age, disabilities, and culture, which may play a significant role in employee performance levels when implementing metaverse technologies. Multiple studies across different industries will help organizations adapt to these new work environments. Finally, it would be vital to carry out studies in multiple industries regarding the metaverse metamorphosis.

In conclusion, the Metaverse era presents significant opportunities and challenges for organizations. To remain competitive, companies need to adopt new practices and performance management systems that are relevant to the current work environment. The systematic literature review highlights the need for new workplace KPIs to measure avatar performance in metaverse environments and the importance of developing robust and validated metrics for measuring avatar performance effectively. A new metaverse performance appraisal management system with the right KPIs is

**Table 1: Avatar-Metaverse-Performance Related Articles Risk of Bias**

| Source/Title                                                                                                                     | Random sequence generation (selection bias) | Allocation concealment (selection bias)              | Blinding of participants and personnel (performance bias)             | Blinding of outcome assessment (detection bias)            | Incomplete outcome data (attrition bias) | Selective reporting (reporting bias) | Other Bias       |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------|--------------------------------------|------------------|
| [64] Organizational communication satisfaction in the virtual workplace                                                          | Randomization by computer-generated list    | Allocation concealment by sealed envelopes           | Participants and personnel were aware of the treatment allocation     | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected     | No evidence of selective reporting   | Publication bias |
| [65] What employees are saying about the future of remote work                                                                   | Randomisation by a coin toss                | Allocation concealment by sealed envelopes           | Participants and personnel were aware of the treatment allocation     | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected   | No evidence of selective reporting   | Language bias    |
| [37] How new ways of working during COVID-19 affect employee well-being via technostress, need for recovery, and work engagement | Randomization by computer-generated list    | Allocation concealment through central randomization | Participants and personnel were unaware of the treatment allocation   | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected     | No evidence of selective reporting   | Measurement bias |
| [66] Key Performance Indicators: A Systematic Literature Review                                                                  | Randomization by computer-generated list    | Allocation concealment by opaque envelopes           | Participants and personnel were not aware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected     | No evidence of selective reporting   | Sampling bias    |
| [67] The relationship between job satisfaction, job performance and employee engagement: An explorative study                    | Randomization by computer-generated list    | Allocation concealment by sealed envelopes           | Participants and personnel were aware of the treatment allocation     | Outcome assessors were not blinded to treatment allocation | Incomplete outcome data were collected   | No evidence of selective reporting   | Selection bias   |
| [68] Visualizing and monitoring effective interactions in online collaborative groups                                            | Randomization by computer-generated list    | Allocation concealment by sealed envelopes           | Participants and personnel were not aware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected     | No evidence of selective reporting   | Processing bias  |
| [69] The impact of training and development on employee job satisfaction, loyalty, and intent to stay in the lodging industry    | Randomization by computer-generated list    | Allocation concealment by opaque envelopes           | Participants and personnel were aware of the treatment allocation     | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected     | No evidence of selective reporting   | Observer bias    |

necessary to ensure an accurate, fair, and comprehensive evaluation of employee performance.

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|                                                                                                                                   |                                          |                                            |                                                                     |                                                            |                                        |                                    |                        |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------|------------------------------------|------------------------|
| [70] Study on performance measurement systems—Measures and Metrics                                                                | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Misclassification bias |
| [71] Work flexibility, job satisfaction, and job performance among Romanian employees                                             | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected | No evidence of selective reporting | Attrition bias         |
| [72] Avatars, people, and metaverses: Foundations for research in virtual worlds                                                  | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Information bias       |
| [73] A proposal for stress management using serious games associated to virtual and augmented reality                             | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Non-response bias      |
| [74] Performance appraisal and performance management: 100 years of progress?                                                     | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Sampling bias          |
| [75] Practica a virtual reality platform for specialized training oriented to improve the productivity                            | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Time-related bias      |
| [7] What Meta employees really think about their company's brutal year                                                            | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected | No evidence of selective reporting | Attrition bias         |
| [60] An evaluation of office productivity measurement                                                                             | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were aware of the treatment allocation   | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected   | No evidence of selective reporting | Selection bias         |
| [76] Driving forces and barriers of Industry 4.0: Do multinational and small and medium-sized companies have equal opportunities? | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected | No evidence of selective reporting | Attrition bias         |



|                                                                                                                               |                                          |                                            |                                                                     |                                                            |                                        |                                    |                       |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------|------------------------------------|-----------------------|
| [33] The effects of self-and external perception of avatars on cognitive task performance in virtual reality                  | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Measurement bias      |
| [77] Crisis management training using discrete event simulation and virtual reality techniques                                | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were aware of the treatment allocation   | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected   | No evidence of selective reporting | Selection bias        |
| [78] Selection of criteria for key performance indicators by the matrix method                                                | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Recall bias           |
| [20] On the Relationship between My Avatar and Myself                                                                         | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Non-response bias     |
| [21] Metaverse                                                                                                                | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Systematic error bias |
| [39] Why The Future Of Work Could Take Place In The Metaverse                                                                 | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Publication bias      |
| [79] Studying the effect of stress factors of on the amount of customers dissatisfaction with using Structural Equation Model | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were aware of the treatment allocation   | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected   | No evidence of selective reporting | Selection bias        |
| [56] Variables of Job Satisfaction: A review study with special reference to LIS Professionals                                | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Measurement bias      |
| [34] Metaverse of madness: Employers and employees at odds over virtual workplaces                                            | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected | No evidence of selective reporting | Attrition bias        |

|                                                                                                                                                         |                                          |                                            |                                                                     |                                                            |                                        |                                    |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------|------------------------------------|-------------------|
| [80] Temporal and Locational Flexibility of Work, Working-Time Fit, and Job Satisfaction                                                                | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Sampling bias     |
| [81] The effect of safety culture on safety performance: Intermediary role of job satisfaction                                                          | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were aware of the treatment allocation   | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected   | No evidence of selective reporting | Selection bias    |
| [82] Virtual reality in neurorehabilitation                                                                                                             | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Information bias  |
| [83] Supporting Teamwork in Industrial Virtual Reality Applications                                                                                     | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were aware of the treatment allocation   | Outcome assessors were not blinded to treatment allocation | Complete outcome data were collected   | No evidence of selective reporting | Selection bias    |
| [84] The challenges of entering the metaverse: An experiment on the effect of extended reality on workload                                              | Randomization by computer-generated list | Allocation concealment by sealed envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Incomplete outcome data were collected | No evidence of selective reporting | Attrition bias    |
| [85] Virtual Human Resource Management in the Metaverse: Immersive Work Environments, Data Visualization Tools and Algorithms, and Behavioral Analytics | Randomization by computer-generated list | Allocation concealment by opaque envelopes | Participants and personnel were unaware of the treatment allocation | Outcome assessors were blinded to treatment allocation     | Complete outcome data were collected   | No evidence of selective reporting | Time-related bias |

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